

Manan N. Ghodasara

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Education

JSS Academy of Technical Education

Bachelor of Engineering in Computer Science

Sept. 2023 – May 2027

Bengaluru, India

- CGPA: 8.6 / 10.0
- Relevant Coursework: Data Structures & Algorithms, OOPS, Operating Systems, Computer Networks, DBMS

Technical Skills

Languages: JavaScript, TypeScript, Python, C++, C
Frontend: React.js, Next.js 14, Vite, Redux Toolkit, Tailwind CSS, Framer Motion, HTML5, CSS3
Backend: Node.js, Express.js, FastAPI, REST APIs, Socket.IO, WebSockets
Databases: MongoDB, PostgreSQL (Prisma ORM), Pinecone Vector DB, SQL
AI / ML: RAG Pipelines, Groq LLM, Gemini, HuggingFace Embeddings, Tesseract.js OCR, OpenCV
DevOps & Tools: AWS (S3), Git, GitHub Actions, Vercel, Render, Prometheus, JWT

Projects

CollabBoard – Real-time Collaborative Whiteboard

[Live](#) [GitHub](#)

React | Node.js | Express | Socket.IO | MongoDB | Redux Toolkit | JWT | Vercel | Render

- Architected a Miro-inspired multi-user whiteboard with **6 drawing tools** and live cursor tracking, achieving **sub-100ms sync latency** across all clients via Socket.IO event streaming with 30ms cursor throttling.
- Optimised pencil rendering using the **Ramer-Douglas-Peucker algorithm**, cutting transmitted path points by **60–80%** and measurably reducing bandwidth under heavy concurrent sketching loads.
- Secured both HTTP and WebSocket layers with JWT; enforced rate limiting (100 req/15 min, 10-attempt lockout) and soft-delete shape history enabling **collaborative undo** across all connected clients.
- Managed global canvas state with **Redux Toolkit**, ensuring consistent shape data and conflict-free undo history across concurrent sessions without race conditions or stale state.

WasteIQ – AI-Powered Restaurant Waste Management SaaS

[Live](#) [GitHub](#)

Next.js 14 | TypeScript | Node.js | Express | PostgreSQL | Prisma | Groq AI | Vercel | Render

- Developed a full-stack SaaS for restaurants with real-time inventory tracking, expiry alerts, and **6-class waste logging** (Spoilage, Expired, Overproduction, Damaged, Quality Issue, Customer Return).
- Embedded a **Groq AI chat assistant** grounded in live restaurant data, **enabling natural-language** inventory queries against the application's current state.
- Built a demand-forecasting dashboard analysing **30 days of sales data** with dynamic waste-pattern charts, PDF report export, and role-based JWT access control for multi-user teams.
- Modelled the entire data layer with **Prisma ORM** on PostgreSQL, enforcing relational integrity across inventory, waste logs, and user roles with end-to-end type-safe query logic in TypeScript.

MindVault – AI Personal Knowledge System (RAG)

[Live](#) [GitHub](#)

React | Vite | Node.js | MongoDB Atlas | Pinecone | HuggingFace | Groq LLM (Llama 3.1 8B) | Tesseract.js | JWT

- Implemented an end-to-end **RAG pipeline**: PDFs chunked into 500-word overlapping segments, embedded to **384-dim vectors** (all-MiniLM-L6-v2), stored in Pinecone, and retrieved via cosine similarity to generate grounded, source-cited answers using Groq LLM (Llama 3.1 8B).
- Achieved **<2s query response** with top-5 chunk retrieval per query; added Tesseract.js OCR for scanned document ingestion and supported unlimited document uploads per authenticated user.
- Designed a **dual-database architecture** – MongoDB Atlas for user accounts and document metadata, Pinecone for persistent vector storage – cleanly separating relational concerns from semantic search.
- Hardened the platform with **JWT + bcrypt** authentication and **express-rate-limit** (10 AI queries/min), preventing abuse while maintaining a smooth experience for authenticated users.

Experience

Software Development & Full-Stack Engineering

Jan 2025 – Present

Independent Technical Projects

Bengaluru, Karnataka

- Built **four applications** across distinct domains — real-time collaboration, AI SaaS, RAG, and a voice-first accessibility assistant (**EchoVision AI**) — working across Node.js, Next.js, and **Python (FastAPI)** backends with a local **OpenCV** computer-vision service.
- Engineered **resilient multi-model LLM integration** with automatic provider fallback (**Gemini primary, Groq fallback**) and context-restricted prompting, keeping AI responses grounded and available even when a provider fails.
- Built a reusable **multimodal extraction stack** — PDF.js for documents and Tesseract.js OCR for scanned images — powering both RAG retrieval and spoken accessibility narration across projects.
- Standardised engineering practice across repos: end-to-end **TypeScript** type safety, JWT-secured APIs with rate limiting, and continuous deployment on **Vercel + Render** with Prometheus monitoring.

Achievements

- **LeetCode**: Solved **210+** problems focused on graphs, dynamic programming, trees, and data structures.
- Participated in multiple AI and full-stack focused hackathons, building end-to-end prototypes under time constraints.
- Completed Apna College's Delta – Full Stack Web Development; Blockchain 101 – Infosys Springboard.